Freight Rate Prediction: Validation Report

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The final model is a linear time model plus LightGBM, trained on about 47,300 clean loads. On four forward-in-time tests it averaged **1.51% MAPE ($36 MAE)** on rows with sound labels, and 3.86% on all rows, because about 1.4% of labels are corrupted and cannot be predicted. Two findings shaped the work: the validation period is entirely in the future, and `quote_signal` is a target leak that stops working in that period.

1. How I split the data

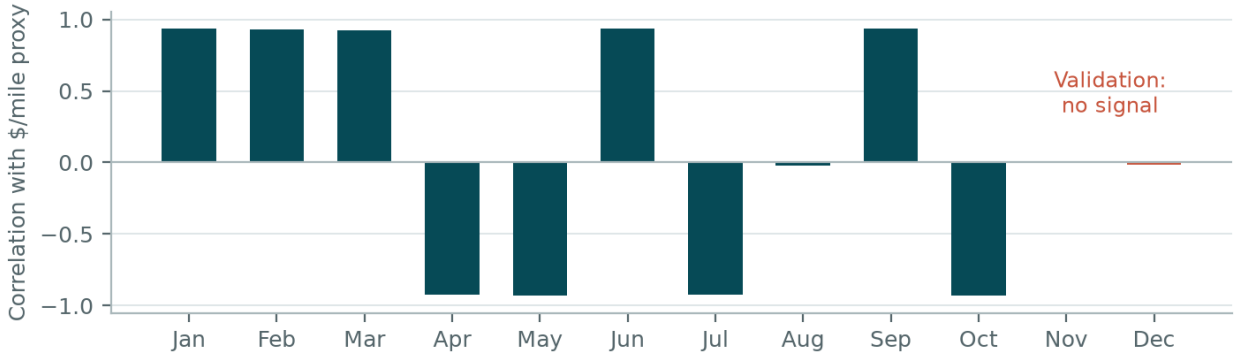
The labelled data runs Jan 1 to Oct 31, 2025. The loads to predict run Nov 1 to Dec 31. So the test that matters is forecasting, and every split trains on the past and predicts the next two months, which is the same horizon as the real task. A random split was rejected: it mixes future days into training and hides time-related failure.

Train through	Predict	Why this fold	MAPE clean	MAE clean	Bias
Apr 30	May-Jun	Same calendar as Nov-Dec: unseen quarter end	1.46%	$37	+0.79%
Jun 30	Jul-Aug	New quarter; quote_signal dead in Aug	1.81%	$43	-1.34%
Jul 31	Aug-Sep	Same calendar as Nov-Dec: unseen quarter end	1.36%	$32	-0.52%
Aug 31	Sep-Oct	Most recent data	1.41%	$33	+0.54%

Bias is mean log(actual / predicted). Corrupted rows are removed inside each fold's training window only; cleaning never sees future labels. For the final fit, all clean rows through Oct 31 are used, since the folds have already chosen the design.

2. Why quote_signal was removed

In Jan-Mar, Jun and Sep this field is the rate per mile plus about 1% noise. In Apr, May, Jul and Oct it is inverted (roughly 4.15 minus rate per mile). In August it is noise, and it is noise in validation too, which can be checked without labels by correlating it with a proxy built from distance and equipment.

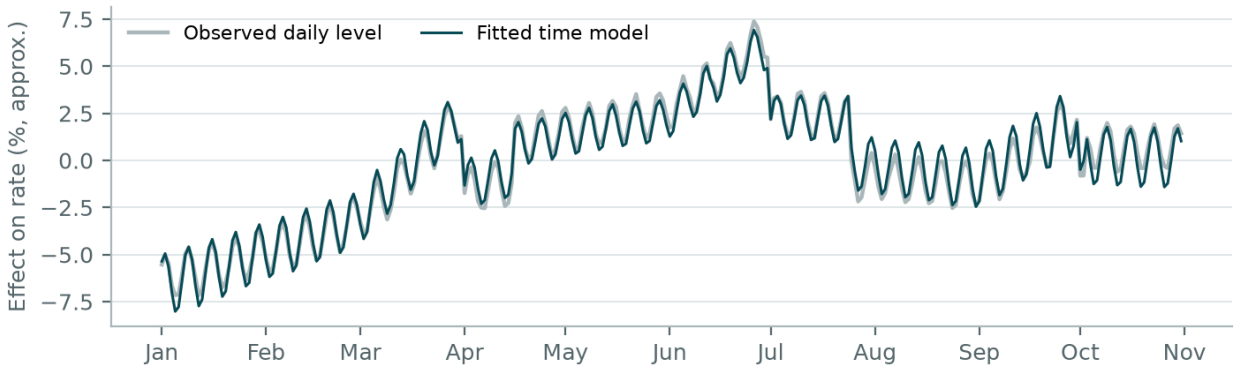


Experiment	MAPE clean
Random 80/20 split, with quote_signal: looks excellent	1.06%
August held out, with quote_signal: the honest result	4.81%
August held out, without quote_signal	1.38%

These three runs use one plain LightGBM. A Sep-Oct holdout alone would not catch the problem, because the field is live then and appears to help.

3. Model and comparison

Trees cannot extrapolate past the last training date, so time is handled by a linear model on log rate per mile: market index (elasticity 0.15), a linear trend (0.6% per 30 days) and a ramp over the last 31 days of each quarter that resets on day one. The ramp differs by equipment, and its size held steady across all three observed quarter ends: Dry Van +2.4%, Reefer +4.7%, Flatbed +7.3%. LightGBM with L1 loss then learns the lane and load shape from distance, weight, equipment and coordinates, on the target with time removed. A small shrunken offset per lane finishes it.



Model (mean of 4 folds)	MAPE clean	MAE clean	MAPE all rows	RMSE all rows
two-stage (final)	1.51%	$36	3.86%	$622
two-stage, no quarter-end ramp	2.45%	$58	4.80%	$625
two-stage, no market index	4.88%	$116	7.28%	$634
LightGBM on date features	2.79%	$66	5.12%	$629
lane median $/mile	6.09%	$147	8.60%	$665

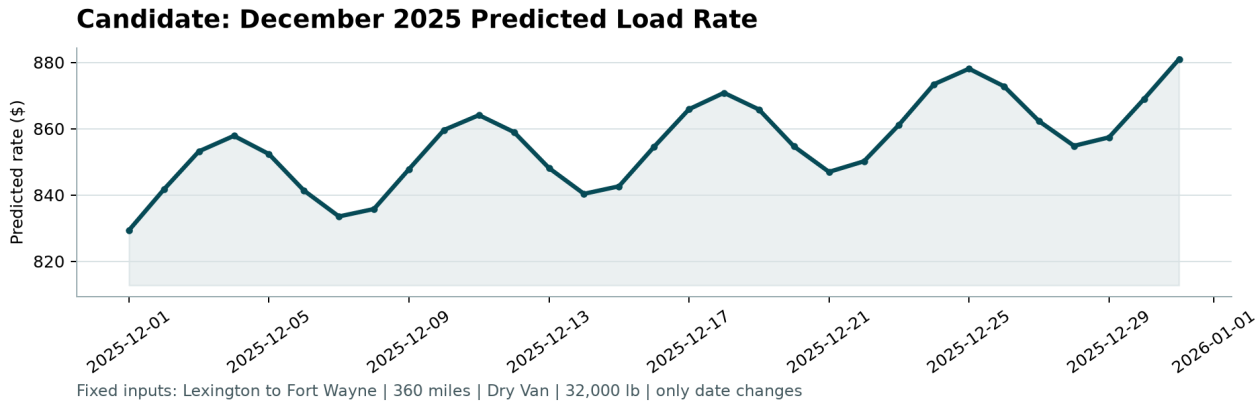
All-row RMSE is dominated by corrupted labels, which is why it barely separates the models. These scores are slightly optimistic, because the same folds chose the feature design and settings.

4. Data quality

Issue	Size	Action
Corrupted rates, about 3.5x and 0.3x true value	677 train rows (1.4%)	Dropped from training by model residual over 0.20 in log terms
Negative weights	292 train, 145 validation	Absolute value; residuals match positive weights
Missing weight	300 train, 165 validation	Train median plus a missing flag
Missing market index	374 train, 249 validation	That date's mean, sd within a day is about 0.025
Capped values	Weight 5,000 to 47,500; distance floor 70	Kept as is
Coordinates distorted (Boston at -69.5 lon, Los Angeles at 28.6 lat)	All cities	Consistent per city, so used as location codes only
Eight cities only in validation	12.1% of validation rows	No city IDs in LightGBM; coordinates carry location

To test unseen cities I hid eight from training and predicted Sep-Oct: 1.70% MAPE on their rows against 1.43% on the rest.

5. Fixed December prediction chart



The chart inputs have no market index, so each day uses the mean market index of that date's validation loads. The weekly wave comes from the index, the climb from the trend and the Dry Van ramp.

6. Assumptions and risks

Holidays: labelled data has no Thanksgiving or Christmas. Jul 4 and Memorial Day show no effect, so I modelled none, and the chart shows no holiday dip or spike. **December ramp:** it rests on three past quarter ends. Removing it would take Dec 31 from $881 to $860 on the chart lane. **Trend:** it is extrapolated two months past the data, and fold bias reached 1.3%. **Market index:** using it assumes it is known on the day of the quote; if it is not, the index would need forecasting in production.